

Automated 3D Stenosis Quantification and Visualization Workflow for Coronary Artery Disease Clinical Decision Support

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Discussion

4.1. Analysis of the Results

- **The Topological Trade-off in MACS-18:** Discuss how the rich anatomical detail of the MACS-18 masks increases the extraction of centerlines (95.62% success) and the number of endpoints, but simultaneously degrades ostium identification accuracy (73.75%). Explain that the skeletonization algorithm gets confused by the dense distal branching.
- **Computational Efficiency and Runtime Discrepancies:** Analyze the execution times. Explain why Block 2 (geometric quantification) is the primary computational bottleneck due to the massive calculation of cross-sectional areas. Justify why the overall pipeline takes significantly longer to run on MACS-18 data compared to ASOCA (precisely because of the higher density of points and distal branches that need to be processed).
- **Algorithmic Verification via Synthetic Models:** Highlight the near-zero error (1.31% max %AS error) achieved on the synthetic models. Explicitly clarify that this small residual error is not a mathematical flaw in the formula, but rather discretization noise generated internally by VMTK when processing voxelized surfaces. This validates the robustness of the underlying mathematical logic.
- **Clinical Usability and Acceptance:** Deep dive into the SUS pilot results (mean score of 70.83). Instead of just stating the final number, break down which specific questions provided the most positive value (e.g., ease of use, intent for frequent clinical use, visual integration) and which were neutral or negative, providing insights into what needs to be polished in the interface.

4.2. Technical and Anatomical Limitations

- **Anatomical Confounders, Tapering, and Area Overestimation:** This is the core limitation of the thesis. Making explicit reference to the newly generated figure that illustrates bifurcations, explain how the area calculation misinterprets the natural widening of a bifurcation as a "healthy" reference zone. This causes a drastic overestimation of stenosis immediately afterward. Firmly conclude that, because of this, the current quantification model lacks sufficient clinical predictive value to act autonomously and strictly requires design adjustments.
- **Rigidity of the Baseline Reference Window:** Analyze the problem with the fixed 10 mm parameter. Discuss the computational complexity of automatically finding two points that truly represent healthy tissue. Relate the fixed-window limitation to the previous point: a rigid 10 mm window is completely blind to whether it is landing on a natural bifurcation or on another calcified plaque.
- **Ostium Detection Fragility:** Acknowledge that the system does not achieve a 100% success rate in ostium detection. Attribute this to potential issues in the source labeling and to highly detailed anatomies complicating the skeletonization endpoints. Admit that there was no material time during this bachelor's thesis to conduct a deep investigation into optimizing this specific detection step (deferring it to future work).
- **Usability Study Sample Size:** Add a brief paragraph recognizing that the SUS test is strictly an exploratory pilot, statistically limited by having been conducted with only 3 expert doctors.

4.3. Future Work

- **4.3.1. Algorithmic and Pipeline Enhancements:**

- Propose the computational optimization and refactoring of the pipeline blocks to make the tool more robust and generalizable.
- **Centerline Resolution Optimization:** Suggest lowering the spatial granularity of centerlines to a coarser spacing (e.g., ~2 mm) to save computational resources where high precision is not strictly needed.
- **Plaque vs. Stenosis:** Clearly differentiate between the presence of plaque and actual hemodynamic stenosis in the algorithmic quantification.
- Investigate robust alternatives for ostium detection to improve current metrics.

- **4.3.2. Clinical Workflow Integration:**

- Propose end-to-end hospital integration by directly connecting this pipeline's inputs to Maren's automatic segmentations/labels and to Eva's patient prioritization system. Explain how the dashboard would function as a rapid hospital triage pop-up to help doctors decide if a patient needs a deeper, immediate evaluation.

- **4.3.3. Addressing Anatomical Limitations: Expert Clinical Proposals:**

(This subsection is entirely dedicated to the hospital's clinical feedback)

- **Dynamic Reference Values:** Solve the 10 mm window issue by allowing cardiologists to dynamically and manually adjust the proximal and distal reference points directly in the UI to bypass bifurcations and anatomical anomalies.
- **Continuous Lumen Visualization & Aggregated Zones:** Shift from showing discrete data points to providing continuous cross-sectional profiles, and highlight aggregated affected zones (e.g., entire regions with $\geq 50\%$ stenosis) to speed up diagnostic reading.
- **Raw Imaging Ground-Truth (The "Clearly" Standard):** Emphasize that to gain clinical trust and emulate highly desired commercial standards like *Clearly*, future versions must integrate Curved Multi-Planar Reformation (CMPR) and cross-sectional views of the real CCTA scans. This ensures absolute explainability against "black-box" models and allows doctors to validate the software's geometric findings against the anatomical reality.